What should your students do?

Tool guide for teachers

Secure the basics

Have terms, facts and formulas at hand

OUR RECOMMENDATION

- **Quiz** ●●○
  Check knowledge
- **H5P (interactive content)** ●●○
  Practise terms
- **Glossary** ●○○
  Record terms
- **AI assistant for course material** AI ●●○
  Look up with source

“Match the ten key terms to their definitions.”

Practise briefly and often, with instant feedback.

Be able to explain

Explain relationships in their own words

OUR RECOMMENDATION

- **Forum** ●○○
  Explain in own words
- **H5P (interactive content)** ●●○
  Match and explain
- **Video with transcript (Panopto)** ●●○
  Explain relationships visually
- **AI tutor in dialogue** AI ●●●
  Explain in conversation

“In five sentences, explain to someone without prior knowledge why …”

Ask for their own words, not memorised definitions.

Practise procedures

Apply a known procedure reliably to new cases

OUR RECOMMENDATION

- **Quiz** ●●○
  Calculation tasks
- **Quiz with STACK** ●●●
  Procedures with changing values
- **Assignment** ●●○
  Show the full calculation
- **AI tutor in dialogue** AI ●●●
  Graded hints while calculating

“Calculate … with your values and show your working.”

Changing values instead of the same task every time.

Break down and compare

Break a case down and compare solutions

OUR RECOMMENDATION

- **Forum** ●○○
  Compare solutions
- **Assignment** ●●○
  Break a case down, define the system
- **Workshop (peer assessment)** ●●●
  Break down another solution
- **Critique an AI answer** AI ●○○
  Find errors and gaps

“Compare two solutions: where do they differ, and why?”

Put two solutions side by side and have the differences named.

Judge with reasons

Check and decide with reasons

OUR RECOMMENDATION

- **Forum** ●○○
  Judge assumptions with reasons
- **Workshop (peer assessment)** ●●●
  Judgement against criteria
- **Assignment** ●●○
  Write a reasoned judgement
- **Critique an AI answer** AI ●○○
  Judge assumptions and sources

“Does this solution hold? Judge it by the three criteria.”

Share the criteria beforehand. No verdict without reasons.

Create something new

Design or develop something of their own

OUR RECOMMENDATION

- **Assignment** ●●○
  Draft, concept, own solution
- **Wiki** ●○○
  Joint product
- **Workshop (peer assessment)** ●●●
  Submit own work
- **Etherpad** ●○○
  Joint draft in real time

“Design your own task on … including a model answer.”

Ask for drafts and give feedback before the final version.

Reflect on their approach

Justify and improve their own approach

OUR RECOMMENDATION

- **Assignment** ●●○
  Learning journal or reflection on their own solution
- **Forum** ●○○
  Share and discuss their approach
- **Feedback and questionnaire** ●○○
  Assess their own progress
- **AI tutor in dialogue** AI ●●●
  Put the next step into words

“Where did you start, and why? What will you do differently next time?”

Often neglected in courses. One reflection question per submission already helps.

Three rules

1. Goal first, then the tool.

2. No tool without a task: only the task turns technology into learning.

3. AI: approved services only, no personal data, grades come from people.

KEY

●○○ low effort
●●○ medium effort
●●● high effort
AI AI tool, status on the back

All tools, examples and background: werkzeuguebersicht.html

Tools from A to Z

Tool guide for teachers

Resources

File and folder ●○○
Provide documents (PDF, slides, scripts)
Good for: Secure the basics

Link, e.g. to a simulation ●○○
Include external pages, simulations and tools

Page and book ●○○
Text, images and video directly in Moodle; the book adds chapters
Good for: Secure the basics · Be able to explain

Video with transcript (Panopto) ●●○
Explain and demonstrate, searchable through chapters and captions
Good for: Secure the basics · Be able to explain

Activities

Assignment ●●○
Submissions with individual grading and feedback
Good for: Practise procedures · Break down and compare · Judge with reasons · Create something new · Reflect on their approach

BigBlueButton ●●○
Video conference in the course with whiteboard and breakout rooms
Good for: Be able to explain · Reflect on their approach

Choice ●○○
One question, several answers, optionally anonymous

Database ●●●
Collect, search and comment entries following a fixed schema
Good for: Be able to explain · Break down and compare · Create something new

Etherpad ●○○
Write one text together in real time
Good for: Be able to explain · Create something new

Feedback and questionnaire ●○○
Questionnaires, also anonymous; the questionnaire module offers more question types
Good for: Reflect on their approach

Forum ●○○
Written discussion, asynchronous
Good for: Be able to explain · Break down and compare · Judge with reasons · Reflect on their approach

Glossary ●○○
Collect and explain terms together
Good for: Secure the basics · Be able to explain

H5P (interactive content) ●●○
Interactive content: fill in the blanks, drag and drop, interactive video
Good for: Secure the basics · Be able to explain

Hot question ●○○
Collect questions, everyone votes for the most important
Good for: Be able to explain · Reflect on their approach

Lesson ●●●
Learning path that branches depending on answers
Good for: Secure the basics · Be able to explain · Practise procedures

Podcast ●●○
Publish audio and video episodes, also by students
Good for: Be able to explain · Create something new

Quiz ●●○
Questions with automatic grading
Good for: Secure the basics · Be able to explain · Practise procedures

Quiz with STACK ●●●
Maths questions with formula input and changing numbers
Good for: Practise procedures

Wiki ●○○
Write and revise texts together
Good for: Be able to explain · Create something new

Workshop (peer assessment) ●●●
Assess each other's submissions against criteria
Good for: Break down and compare · Judge with reasons · Create something new · Reflect on their approach

AI tools

AI assistant for course material
AI · pilot ●●○
Answers questions from the stored material only and cites the passage
Good for: Secure the basics · Be able to explain

AI feedback on the solution path
AI · concept ●●●
Handwritten path is recognised and checked against criteria
Good for: Practise procedures · Break down and compare

AI tutor in dialogue AI · in use ●●●
Gives no solution but asks step by step
Good for: Be able to explain · Practise procedures · Break down and compare · Reflect on their approach

Critique an AI answer AI · method ●○○
A pre-generated AI answer becomes the object of review
Good for: Break down and compare · Judge with reasons · Reflect on their approach

Only use AI services approved by the university. Do not put personal data or exam work into external services. Grades are given by the teacher, not by the AI. The recommendations are a reasoned professional judgement, not a measurement.

Foundations: Lorin W. Anderson und David R. Krathwohl (Hrsg.): A Taxonomy for Learning, Teaching, and Assessing. New York 2001. · Punya Mishra und Matthew J. Koehler: Technological Pedagogical Content Knowledge. Teachers College Record 108, Nr. 6 (2006): 1017–1054. · KIViAzubi (FKZ 13HAW013FQ), Phasenabschlussbericht „Vier Blicke auf einen Kurs“, Schicht A, HSD/ZWEK 2026. · Moodle Tool Guide: Idee Joyce Seitzinger, erweitert von Gavin Henrick, deutsche Übertragung Ralf Hilgenstock, Susanne Gebauer, Gerald Hartwig; Anpassung FH Düsseldorf FB 6 von Christian-Maximilian Steier (2013), CC BY-NC-SA.

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